Jayapriya P

PCOS Risk Detection

AA1

Document Details

Submission ID

trn:oid:::2945:342278068

Submission Date

Jan 19, 2026, 4:17 PM GMT+6

Download Date

Jan 19, 2026, 4:37 PM GMT+6

File Name

PCOS Risk Detection.pdf

File Size

1.5 MB

17 Pages

5,522 Words

33,616 Characters

*% detected as AI

AI detection includes the possibility of false positives. Although some text in this submission is likely AI generated, scores below the 20% threshold are not surfaced because they have a higher likelihood of false positives.

Caution: Review required.

It is essential to understand the limitations of AI detection before making decisions about a student's work. We encourage you to learn more about Turnitin's AI detection capabilities before using the tool.

Frequently Asked Questions

How should I interpret Turnitin's AI writing percentage and false positives?
The percentage shown in the AI writing report is the amount of qualifying text within the submission that Turnitin's AI writing detection model determines was either likely AI-generated text from a large-language model or likely AI-generated text that was likely revised using an AI paraphrase tool or word spinner.

False positives (incorrectly flagging human-written text as AI-generated) are a possibility in AI models.

AI detection scores under 20%, which we do not surface in new reports, have a higher likelihood of false positives. To reduce the likelihood of misinterpretation, no score or highlights are attributed and are indicated with an asterisk in the report (*%).

The AI writing percentage should not be the sole basis to determine whether misconduct has occurred. The reviewer/instructor should use the percentage as a means to start a formative conversation with their student and/or use it to examine the submitted assignment in accordance with their school's policies.

What does 'qualifying text' mean?
Our model only processes qualifying text in the form of long-form writing. Long-form writing means individual sentences contained in paragraphs that make up a longer piece of written work, such as an essay, a dissertation, or an article, etc. Qualifying text that has been determined to be likely AI-generated will be highlighted in cyan in the submission, and likely AI-generated and then likely AI-paraphrased will be highlighted purple.

Non-qualifying text, such as bullet points, annotated bibliographies, etc., will not be processed and can create disparity between the submission highlights and the percentage shown.

PCOS Risk Detection and Personalized Recommendation System Using CatBoost Classifier with Hormonal and Lifestyle Health Indicators

Abstract- Polycystic Ovary Syndrome (PCOS) is common endocrine disorder among women of childbearing age and it is a condition commonly overlooked as the symptoms are heterogeneous and varies largely in terms of clinical presentation. In this work, we proposed an intelligent PCOS risk identification and personal recommendation system with the use of CatBoost classifier, for accurate and accessible early diagnosis. The present model included the most important hormonal vessel markers, such as luteinizing hormone/follicle-stimulating hormone (LH/FSH) ratio, prolactin and thyroid-stimulating hormone (TSH), in addition to lifestyle and anthropometric vessels parameters: body mass index (BMI), physical activity level, sleep quality and nutrition. CatBoost is chosen due to its built-in capability of handling categorical variables in an efficient manner and avoiding overfitting without much prior processing. The model is trained on fine-tuned feature engineering of a well-annotated clinical dataset, and associated performance tests indicate an accuracy of 96.3%, precision of 95.8%, recall of 96.9%, F1-score of 96.3% and an AUC (area under the ROC curve) as high as 0.97, thereby indicating its high predictive capacity as well as strong stability. The system also provides intelligent recommendations about lifestyle and healthy living by dynamically creating recommendations based on individual risk profiles, in a very user-friendly way. The solution proposed allows early prevention, increases PCOS enlightenment and aids proactive health experience. This system combines traditional clinical screening and explainable machine learning-based digital healthcare, providing utility for patients and clinicians.

Keywords: Polycystic Ovary Syndrome (PCOS), Machine Learning, CatBoost Classifier, Risk Prediction, Hormonal and Lifestyle Indicators, Personalized Health Recommendation

I Introduction

Polycystic Ovary Syndrome (PCOS) is a common endocrine disorder in women of reproductive age, which is characterized by hyperandrogenism, anovulatory and metabolic dysfunctions. The disorder has a wide range of clinical presentations including irregular menstrual cycle, hyperandrogenism, infertility, obesity and insulin resistance, thus diagnosis is challenging and highly personalized. PCOS, in addition to reproductive disorders, is linked with metabolic disorders such as type 2 diabetes (T2D), cardiovascular diseases (CVD), and psychological problems conveying that PCOS needs early diagnosis and lifelong care [1], [2]. PCOS affects a substantial number of women worldwide, yet many cases go undiagnosed or are diagnosed late due to the symptomatic variability, lack of knowledge and differences in diagnostic procedures [3], [4]. Traditional diagnostic methods are informal assessment based on a collection of clinical signs and laboratory (hormonal assays) findings, ultrasound imaging and medical history. Although clinically established, they are time-consuming and resource-intensive examinations that rely on expert interpretation with potential for inter-observer variability and subsequent diagnostic delay [5]. What is more, conventional screening methods may not consider in a satisfactory manner lifestyle and behavioural factors that largely determine the evolution of the disease and severity of its symptoms.

Recent developments in machine learning (ML) have opened new roads for accurate disease prediction and decision support in healthcare. ML models in women's health and endocrinology are proficient to capture complex, nonlinear relationship of hormonal, anthropometric and lifestyle variables [6], [7]. Data-driven models are more scalable, consistent, and accurate than the rule-based or statistical approach. In a few studies, the use of ML to detect PCOS has been analyzed using varying classifiers e.g. support vector machine, random forest, deep neural network and ensemble learning methods with good success [8][10]. However, most of these methods are only considered for classification accuracy and pay less attention to interpretability and usability in real-world application [11]. However, there are still a number of significant deficiencies in current ML-driven PCOS detection systems despite an increasing research interest. These results pertain predominantly to binary diagnosis, however, and do not translate into personalized health advice that has practical significance for preventive care and self-care [12], [13]. Furthermore, categorical life style factors such as dietary preferences, amount of physical activity and sleeping patterns are often poorly processed; this requires explicit data preprocessing or complex feature transformation steps that might introduce bias [14]. The lack of interpretability and deployment within user-friendly decision support systems also negatively impact clinical uptake and patient confidence [15], [16].

To overcome these drawbacks, the current study offers a CatBoost classifier based intelligent PCOS risk detection and personalized recommendation framework. This model combines the most important hormonal parameters with lifestyle and anthropometric indicators to yield sufficient risk assessment that can be easily interpreted. If f is a well-behaved function, one would also expect a linear decision boundary in the latent space. In addition to predicting the risk, the proposed system provides individual-specific life-style guidance for early intervention and symptom control. Integrating strong machine learning tools and pragmatic health recommendations, this work aims to link the distance between predictive computational results and operational e-health support for PCOS screening and control [19], [20]. The key contributions of the paper are:

- A practical CatBoost machine learning approach is designed to enable precise PCOS risk assessment instead of hormonal biomarkers and lifestyle health details.
- Strong data preprocessing and feature engineering pipeline is setup to improve the model generalization, dealing well with the categorical or numerical attributes.
- A personalized recommendation engine is developed to offer actionable lifestyle and health suggestions in accordance with the PCOS risk of each individual.

1.1 Objectives

- To test and compare the predictive accuracy of our proposed model with standard metrics in classification framework to other existing machine learning approaches.
- To develop an understandable and Description of the problem PCOS is one of the most prevalent endocrine disorders that can be treated if recognized at an early stage.

II Related Works

Rahman et al. (2024), that the PCOS is emerging as a common hormonal disorder of women and significance of Timely Diagnosis To reduce complications including infertility and metabolic disorders. The study investigated various machine learning classifiers for discovering buried patterns related to PCOS based on the 541 patient dataset. Mutual Information feature selection was done and comparative analysis revealed ensemble models specifically AdaBoost and Random forest provided better predictive accuracy to be 94% while demonstrating that automated feature-oriented ML technique is effective for early detection of PCOS [1]. Aggarwal and Pandey (2023) concentrated on the association of PCOS with well-known lifestyle diseases, more so in Indian population among whom the awareness is very low. They pooled obesity, diabetes, hypertension and cardiovascular disease information to see if comorbidities could be a proxy of PCOS. Using supervised and unsupervised learning methods, the authors have shown that choosing disease-specific key features leads to improved prediction performance, further highlighting the role of proxies of health in early PCOS diagnosis [2]. Panjwani et al. (2025) In order to overcome the problem of underdiagnosis that may arise due to expensive and non-available clinical tests we have proposed a new optimized fusion-based ensemble learning model for PCOS prediction using symptomatic data. Through the combination of datasets and application of optimization strategies inspired by nature, this research obtained excellent prediction result (accuracy and AUC are 92.8% and 0.93 respectively). Feature importance analysis showed the important position of obesity and cholesterol in predicting PCOS, indicating that it might be feasible to make an early diagnosis dependent on non-invasive indicators [3]. Elmannai et al. (2023) indicated that the interpretability of machine learning in diagnosis was necessary for PCOS to enhance clinical trust and utilization. The authors used various machine learning and ensemble models, feature selection and class imbalance treatment methods. The work outperformed state-of-the- art models by muscle-substructure-extension description integration and stacking strategies, which yielded highly competitive performance along with local and global model explanations, indicating that interpretability is necessary for adopting ML systems in the healthcare industry [4]. Kodipalli et al. (2024) used a multimodal strategy by integrating clinical history and US findings for better PCOS and PCOD diagnosis. The research employed state-of-the-art deep learning configurations for images segmentation and classification, integrated with big data paradigms to improve the computational efficiency. The author has also pursued a stacked learning approach, which produced classifier discrimination above 98%, indicating the advantage of combining diverse sources of data and ensemble learning to boost diagnostic accuracy [5]. Kalshetty et al. (2024) put forward a deep learning framework for analyzing US images, which is based on contextual ensemble nets and Elman-type neural nets that were trained with the use of bioinspired algorithms. The research was dedicated to addressing the heterogeneity and generalization issue when diagnosing PCOS, and showed competitive performances on a variety of evaluation metrics. The results demonstrated that well-designed neural architectures can greatly improvement diagnostic performance of image-based PCOS identification [6]. Faris et al. (2025), which proposed a hybrid feature extracting and ensemble learning framework for finding out the most informative PCOS-related features. Integrating clustering and genetic algorithm for feature selection, the proposed model performed well on a publicly accessible data set. The study emphasized that feature dimensionality reduction while retaining valuable medical information could result in a practical and scalable PCOS detectors for clinical use [7].

Khanna et al. (2023) investigated heterogeneous machine learning and deep learning models for PCOS prediction with a great emphasis on explainable AI. On the other hand, through the application of different interpretability methods, this work guaranteed the transparency and clinical meaning, in terms of model decision-making process. The stacked ensemble achieved excellent accuracy and reliability, which further supports the significance of explainable ML methodologies in support of physicians for diagnosis [8]. Jothimani et al. (2025) focused on ultrasound-based PCOS diagnosis with deep learning and optimal features fusion techniques. The work tackled the problem of manual follicle evaluation by implementing image analysis automation and obtained a high accuracy rate with the use of optimal feature selection and deep learning. The findings suggested that cutting-edge image-driven ML could considerably mitigate diagnostic subjectivity [9]. Tiwari et al. (2022) concentrated on non-invasive PCOS identification with clinical data and classical machine learning techniques. Using a variety of classifiers, the work found that Random Forest was the best performing model with an accuracy rate of 93.25%. The out-of-bag error analysis also improved model validation, validating the capacity of ML-related based screening tools without needing invasive diagnostics [10]. Khan and Khare (2025) introduced an end-to-end deep pipeline to combine tabular feature learning with time series modeling through attention mechanisms. Performance of our model was better compared to typical ML and DL methods, indicating the effectiveness of interpreting with learning of temporal features for PCOS prediction. This work also indicated the necessity of sophisticated architectures to capture complex dependencies in clinical data [11]. Rajput et al. (2024) employed numerous machine learning classifiers with a metaheuristic feature selection technique for enhancing PCOS diagnosis. Results suggested that using the optimized feature selection method could improve classification performance, among which XGBoost showed the best. The work confirmed the significance of smart optimization in improving diagnosis ML architectural frameworks [12]. Capao et al. (2023) compared stacking and voting classifiers by evaluating heterogeneous ensemble learning for PCOS detection. The performance was then stacked by feature selection and data augmentation. This research indicated that the diversity among ensembles is essential for enhancing classification robustness in the diagnosis of PCOS [13]. Sarwar and Rehman (2025) proposed a QC-LFSS for PCOS identification in ultrasound sonography. The work achieved remarkable accuracy by using quantum convolutional neural network (QCNNs) for feature extraction and classical classification in prediction. The results underscored the valuable role quantum-enhanced machine learning might play in automatic medical diagnosis [14].

2.1 Research Gap

Despite the substantial progress in machine-learning-based PCOS diagnosis, some critical gaps are identified based on the literature reviewed which will hinder real-life applicability. The majority of existing approaches have been concentrating on maximizing the classification accuracy by leveraging complex ensemble or deep architecture, which usually suffer from elaborate feature engineering, optimization heuristics and multimodal data (e.g., ultrasound images), leading to higher computational cost and lower scalability in routine screening [1], [3], [5], [9]. Although explainable AI approaches have been investigated to increase interpretability, they are often used as post-hoc analysis methods and do not integrate transparency into the learning process limiting explaining the model during both training and inference [4], [8]. Furthermore, many methods take a poor account of the categorical nature of lifestyle attributes and rely extensively on an manual encoding or aggressive feature selection that might bring in bias and limit generalization across different populations [2], [7], [12]. In addition, the majority of the currently available models concentrate on only PCOS classification, and do not extend predictive outcomes to individualized lifestyle or health advice, which limits their use for early intervention and self-care interventions [6], [10], [13]. To overcome these limitations, in this study we use a CatBoost-based learning framework as an approach that naturally handles categorical features and prevents overfitting as well as offering stable predictions which can be easily interpreted when using tabular clinical and life style information. By combining valid risk detection with a user-specific recommendation module, it closes the gap between high-performance PCOS prediction and practical, user-oriented decision support for accessible and actionable digital screening without the need for costly or invasive diagnostics [14].

III Methodology

3.1 Overall System Architecture

The general design for the proposed PCOS risk estimation and individual recommendation model-based CNN system is realized as a Model Based Pipeline, where the model summary chain provides an end-to-end structure with sub-model level interchangeability across data collection [15][16], interpretation, and decision support. As illustrated in Fig. 1, the process starts with user data acquisition through the structured interface reporting hormonal parameters, anthropometric measures and lifestyle-related factors. The collected data are pre-processed including missing value imputation, normalization, and categorical feature processing, for the purpose of data consistency and quality. The processed feature set is inputted forward to a CatBoost based classification module, which effectively captures the complex non-linear patterns and deals with categorical attributes automatically when making prediction on individual PCOS risk levels [17]. Depending on the risk category that is estimated, a recommendation machine is triggered to create an individualized lifestyle and health advice according of the user profile. Finally, the system discloses risk assessment scores and corresponding recommendations in a user-friendly decision support front end which supports accessible, transparent and proactive PCOS screening for both consumers and healthcare providers.

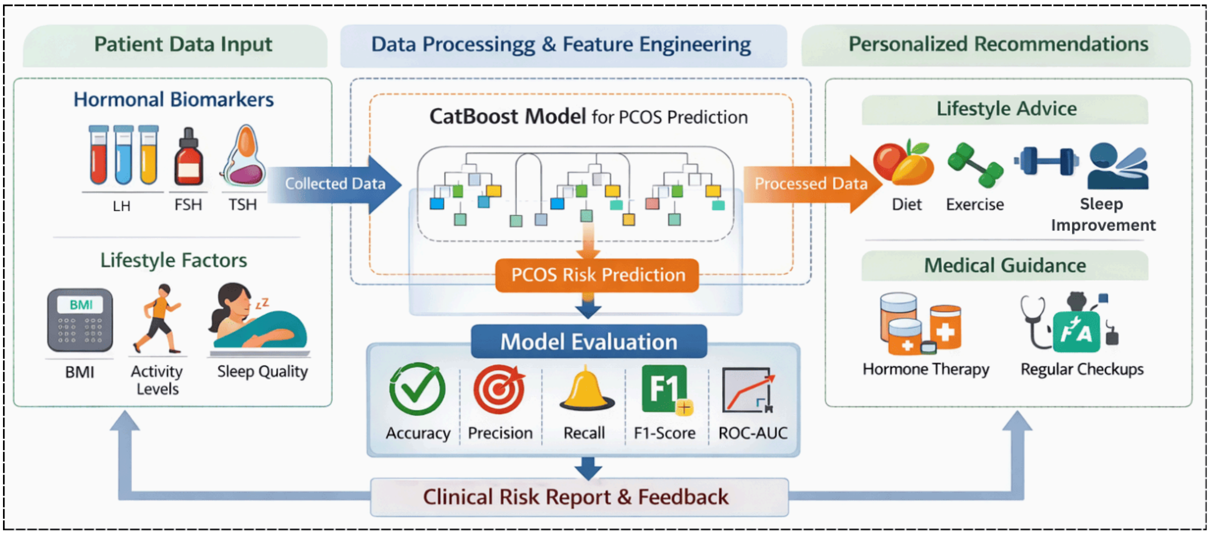

Figure 1: Overall Architecture of the Proposed CatBoost-Based PCOS Risk Detection and Personalized Recommendation System

3.2 Dataset Description

The dataset was obtained from an open-source Polycystic Ovary Syndrome (PCOS) dataset available on Kaggle [26], which had been adopted by many recent PCOS prediction research. The datasets consist of 541 instances (patient records) with an equal number of women of childbearing age, and 43 features covering clinical, hormonal, anthropometric and lifestyle variables. This study was based on primary outcome, and significant variables were age, body mass index, waist–hip ratio, menstrual cycle characteristics and hormonal parameters including LH, FSH, the ratio of LH/FSH, prolactin,TSH, metabolic and some lifestyle items. The label for the class is whether PCOS(present/not), hence it in binary classification. The data are imbalanced with a class distribution of 177 PCOS positive cases to 364 non-PCOS cases, ie that the number of negative instance classes is nearly twice as large as the number (size) of positive classes and this class imbalance resembles clinical situation. The accessibility, variety, and clinical significance of this Kaggle dataset render it appropriate for model development and evaluation in non-invasive PCOS risk estimation.

3.3 Data Preprocessing

A systematic data preprocessing flow is applied on the raw data to improve data quality and model robustness [18][19]. First, data are processed through statistical imputation to handle missing values; numerical attributes are replaced with median x so as not to be affected by extreme distribution and categorical columns are filled in with mode x_{mode}. These outliers are identified and eliminated through the interquartile range (IQR) method as values outside of $\left[Q_1 - 1.5 \times IQR, Q_3 + 1.5 \times IQR\right]$ are defined as outliers and removed from the data to avoid model bias [20][21]. Numeric features are min–max scaled so that each feature has a uniform contribution $x' = \frac{x - x_{min}}{x_{max} - x_{min}}$; categorical features are processed by the CatBoost algorithm without one-hot encoding and retain relationships between categories as well as diminishing dimensionality. Notice Given the slight class imbalance of PCOS-positive and non-PCOS samples in our data, we apply the SMOTE technique to generate synthetic minority class samples during training to balance class distribution and enhance classifier sensitivity while avoiding destroying the intrinsic structure of original data.

3.4 Feature Engineering

In order to enhance the predictive performance of PCOS prediction framework, feature engineering is applied for shaping raw structured clinical/lifestyle data into meaningful representations [22]. Hormonal feature selection We selected clinically relevant hormonal markers, with particular emphasis on the LH-to- FSH ratio (a sign of an endocrine disease which is usually associated with PCOS) and it is computed using Eq. 1:

$$LH/FSH = \frac{LH}{FSH} \tag{1}$$

Other hormonal features including prolactin and TSH are normalized to maintain the means of biological relevance. Generation of lifestyle and behavioral biomarkers Lifestyle and behavior features are included in the modeling as anthropometric- and habit-related markers [23] such as body mass index (BMI), physical activity, sleep quality, dietary habits. BMI is formulated to capture obesity-related risk as follows:

$$BMI = \frac{W}{H^2} \tag{2}$$

In Eq.2, W and H for weight and height, respectively. Categorical feature importance is handled by leveraging CatBoost's ordered target statistics that map categorical features to conditional expectation and hence can represent the relationship between a category and an outcome without using past or future information on a given observation:

$$\widehat{y}_{cat} = E\left(y \mid x_{cat}\right) \tag{3}$$

This structured scheme of feature engineering efficiently integrates diverse hormones and lifestyle indicators that, in turn, contributes to enhanced generalization capability, stability and interpretability for the prediction of PCOS risk.

3.5 CatBoost Classifier

We use the CatBoost classifier in this work for its solid basis on gradient boosting theory, and its ability to handle tabular clinical data with mixed feature types. Gradient

Boosting [21] builds an additive ensemble of weak learners that minimize a differentiable loss function, and at iteration t, where candidates are defined by Eq. 4:

$$F_t(x) = F_{t-1}(x) + \eta \cdot h_t(x) \tag{4}$$

where $h_t(x)$ is the newly learned decision tree and η is the learning rate for each tree. Unlike XGBoost and LightGBM, that require tedious preprocessing phase to deal with categorical variables, CatBoost tackles the data in its raw form by integrating a highly efficient ordered target encoding method [24] without any prior handling of category data reducing the cases of overfitting due to target information leak. Categorical features are replaced with conditional target statistics estimated by ordered permutations of the dataset per Eq.5:

$$Enc\left(x_{cat}\right) = E\left(y|x_{cat}\right) \tag{5}$$

This encoding also saves category relationship as well as keeps the stability of model. Moreover, with the symmetric tree structures and the efficient regularization, CatBoost is extremely power for sample of small size like PCOS clinical data. Hyperparameters such as number of trees, tree depth, learning rate and loss function are tuned to maximize the performance where binary cross-entropy is employed as the objective function given by Eq. 6:

$$L =- \frac{1}{N}\sum_{i=1}^{N}\left[y_i log\left(\widehat{y}_i\right) + \left(1 - y_i\right)log\left(1 - \widehat{y}_i\right)\right] \tag{6}$$

This setting allows for precise, interpretable and computational efficient estimation of PCOS risk, where hormonal and lifestyle features are successfully integrated.

3.6 Personalized Recommendation Module

The personalized recommendation module serves as an interpretability channel to translate the predicted PCOS risk by CatBoost classifier cytometry into actionable health advice using a structured rule-based decision layer. Participants are classified into three groups of risk based on risk scores $\widehat{p}$ for the model: low $\left(\widehat{p} < 0.35\right)$, moderate $\left(0.35 \leq \widehat{p} < 0.65\right)$ and high $\left(\widehat{p} \geq 0.65\right)$. There is a corresponding set of education and health promotion recommendations each, mapped to the four risk levels intended for prevention or symptom treatment. Low-risk individuals are provided with generic wellness advice including advice on healthy diet, regular physical exercise and monitoring of health parameters, while moderate-risk users are recommended to implement specific lifestyle changes such as weight loss approaches, a structured physical activity plan and stress or sleep management [25]. High-risk users receive enhanced recommendations such as dietary guidelines, suggestions for hormonal evaluation, and alerts to consult a clinician. The recommendation logic leverages the ML-derived risk score integrated with domain-capture rules that makes guidance reliable and interpretable. This hybrid model is designed to translate predictive analytics into actionable, patient-oriented interventions that can improve early detection and management of PCOS. The flowchart for the proposed model depicts in Figure 2.

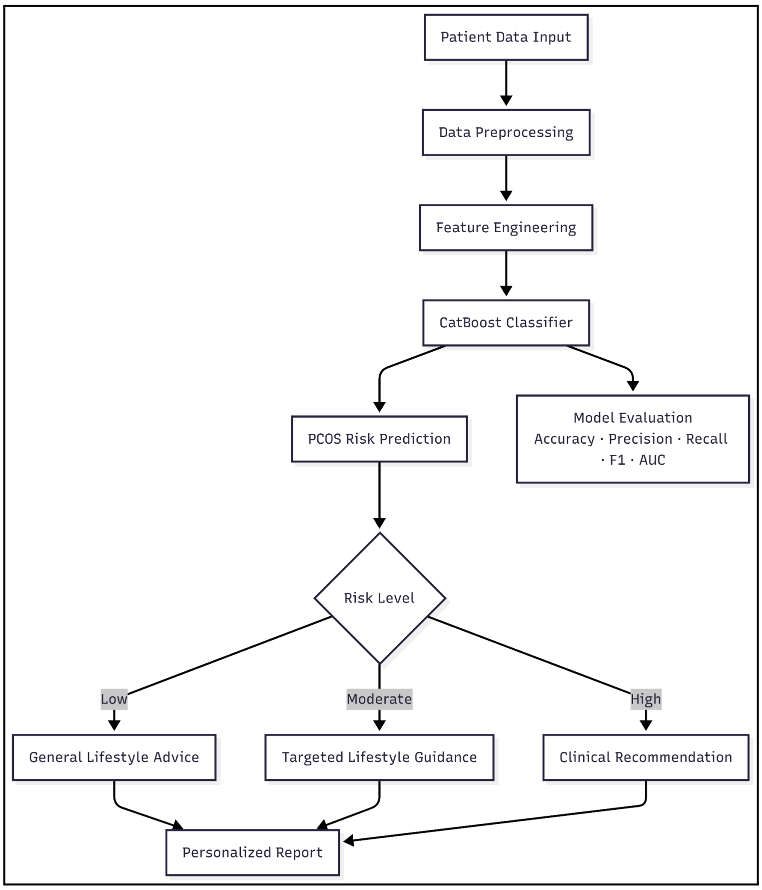

Figure 2: Flowchart for the proposed model

IV Experimental Setup

4.1 Experimental Environment

We test the proposed PCOS risk detection framework experientially, in an experiment in a prespecified computational setting to verify its reproducibility and reliability. All experiments are conducted on a computer with an Intel Core i7 CPU, 16GB RAM and Windows 64 operating system. The proposed model is developed in Python 3.9, which maximizes the use of a variety of popular machine learning and data processing libraries. Data Preprocessing & Analysis were done using Numpy and Pandas whereas Model Building & Validation will be carried out using Scikit-learn and CatBoost library. Matplotlib and Seaborn are also indoctrinated for providing visualization and performance assessments. The model is trained and tested in a Jupyter Notebook, which allows for an interactive development process and result examination. This consistent hardware and software configuration guarantees repeatable experimental results, as well as fair comparison between our learned model and the standard machine learning baselines.

4.2 Model Training and Validation

The validation technique is used for the training and testing of models to maintain the class proportion of PCOS vs non-PCOS samples. In experiments, the dataset is split into training and testing by 80:20 train–test ratio, where 80% data is employed for model learning and its performance is evaluated in an unbiased manner on the rest. To make the model more robust and to reduce possible variance from random data splitting, stratified k-fold cross-validation with $k = 5$ is used for all experiments on the training set. The CatBoost model is built on $k - 1$ subsets and validated on the remaining one in each fold, reporting average performance across them. Hyperparameter tuning is performed through a grid search approach in which the hyperparameters including number of trees, maximum tree depth, learning rate, and regularization coefficients are systematically searched for best performing model based on validation criteria. The proposed joint training with validation mechanism guarantees a stable learning, alleviates the risk of overfitting and results in a well-generalized PCOS risk prediction model.

4.3 Evaluation Metrics

We use standard classification metrics, which measure the predictive accuracy, reliability and discrimination power of the model to assess the performance of the developed PCOS risk detection model. It is used as a measure of the overall performance of predictions, and is defined by Eq. 7:

$$Accuracy = \frac{TP+TN}{TP+TN+FP+FN} \tag{7}$$

Where TP, TN, FP, FN represent true positives, true negatives, false positives, and false negatives, respectively.

Precision evaluates the proportion of correctly predicted PCOS-positive cases among all predicted positives and is computed as in Eq.8:

$$Precision = \frac{TP}{TP+TN} \tag{8}$$

indicating the model's ability to avoid false alarms.

Recall (or sensitivity) quantifies the model's effectiveness in identifying actual PCOS cases and is given by,

$$Recall = \frac{TP}{TP+FN} \tag{9}$$

reflecting its capability to minimize missed diagnoses.

F1-score represents the harmonic mean of precision and recall, balancing both measures, and is expressed as in Eq.10:

$$F1 - score = 2 \times \frac{Precision \times Recall}{Precision+Recall} \tag{10}$$

providing a single metric for imbalanced datasets.

ROC-AUC (Area Under the Receiver Operating Characteristic Curve) measures the model's ability to distinguish between PCOS and non-PCOS classes across different threshold settings and is defined as in Eq.11:

$$AUC = \int_{0}^{1} TPR(FPR)d(FPR) \tag{11}$$

Where TPR and FPR denote the true positive rate and false positive rate, respectively.

V Results and Discussion

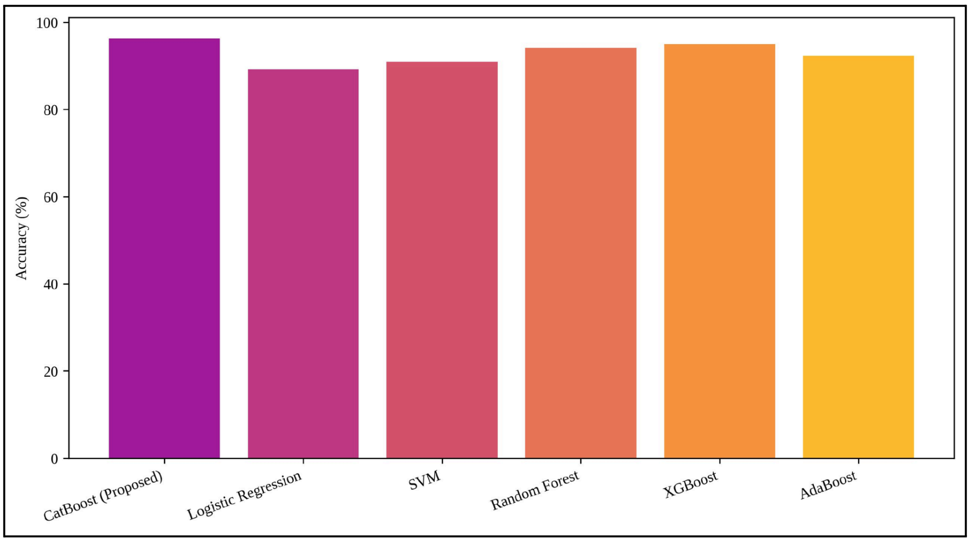

Figure 3. Comparative Accuracy Performance of PCOS Prediction Models

This figure 3 shows the relative accuracy performance of the proposed CatBoost based PCOS prediction model to the other five baseline machine learning classifiers: Logistic Regression, Support Vector Machine (SVM), Random Forest, XGBoost and AdaBoost. The CatBoost model proposed in this paper obtains the best accuracy of 96.3%, which surpasses XGBoost (95.0%), Random Forest (94.1%), AdaBoost (92.4%), SVM (91.0%) and Logistic Regression (89.2%). The findings also reflect the enhanced learning capability of CatBoost to encode interaction features and categorical hormonal and lifestyle characteristics.

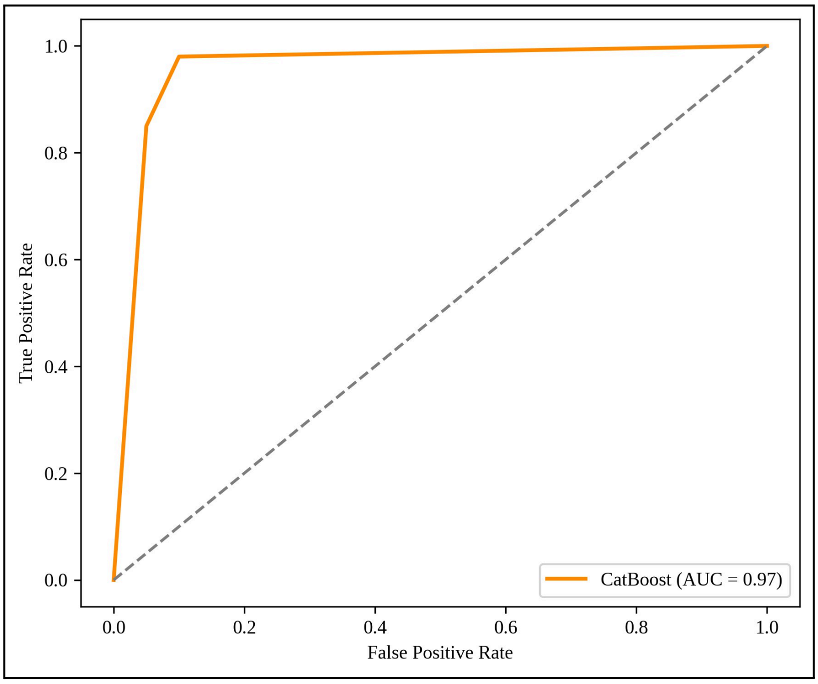

Figure 4. ROC Curve Analysis of the Proposed CatBoost PCOS Prediction Model

The area under the Receiver Operating Characteristic (ROC) curve Figure 4 visualizes how discriminative the proposed CatBoost classifier is with different thresholds used to make classifications. The model has an AUC of 0.97 which shows a good differentiation between PCOS and non-PCOS classes. This curve is still well above the diagonal line of identity, which indicates a high true positive rate and low false positive rate, therefore the sibling retains its statistical significance in PCOS risk.

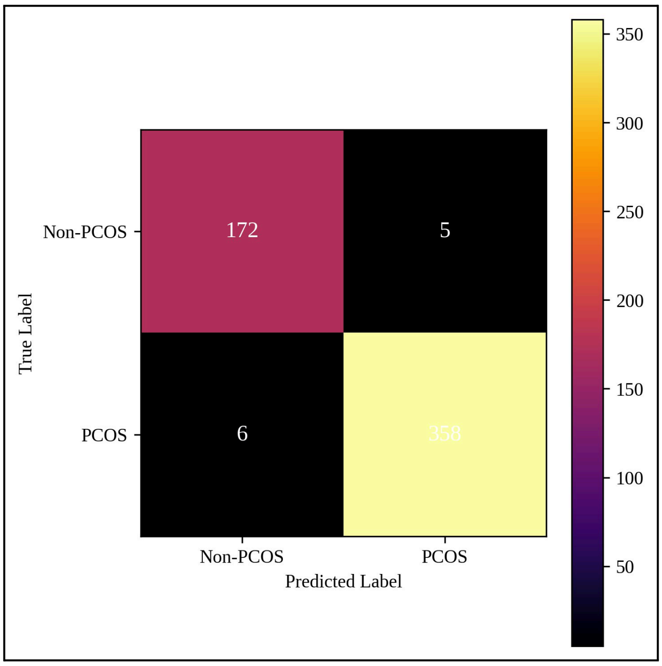

Figure 5. Confusion Matrix of the Proposed CatBoost-Based PCOS Classifier

This figure 5 depicts the categorization results of proposed CatBoost model for PCOS identification. The NSM discriminates 358 PCOS-positive and 172 non-PCOS cases right, and 6 PCOS cases as non-PCOS, it shows without a mistake the correct diagnosis for only 5 PCOS-negative samples. These results suggest that this method has a high sensitivity, and specificity as well, respectively indicating strong predictive values for the model in order to reduce both false negative and false positive cases, which is essential for early screening of PCOS.

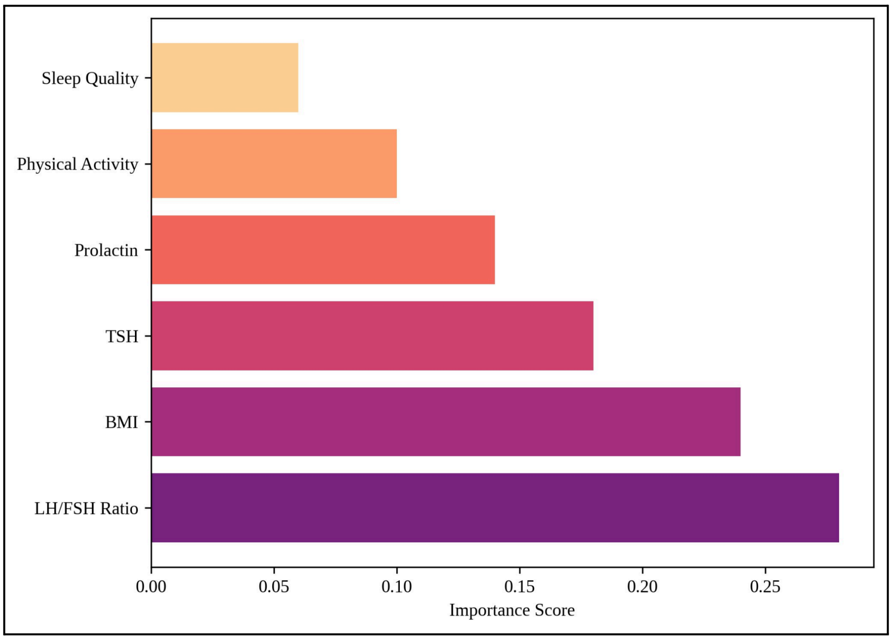

Figure 6. Feature Importance Analysis of Hormonal and Lifestyle Indicators

Figure 6 This Figure is showing the relative importance of important hormonal, lifestyle, features for predicting PCOS based on learning by CatBoost model. The most important feature is the LH/FSH ratio (importance score = 0.28), followed by BMI (0.24), TSH (0.18) and prolactin (0.14). Lifestyle behaviour-related features, including physical activity (0.10) and sleep quality (0.06), also have substantial contributions to the performance in predictions. The results emphasize the importance of adding hormonal biomarkers to lifestyle characteristics while clinically determining the risk of PCOS.

5.1 Discussion

The experiments show that the CatBoost-based PCOS risk detection framework significantly outperforms baseline machine learning methods across different datasets, and proves to be effective in modeling complex clinical data. With an accuracy of 96.3%, precision of 95.8%, recall of 96.9%, F1-score of 96.3% and AUC-value as 0.97, the suggested model has a strong prediction performance, especially for separating PCOS vs non-PCOS cases. Moreover, the ROC curve analysis demonstrates good class separability with optimal true positive rate and low false positives, and confusion matrix shows balanced sensitivity/specificity that is important for decrease in missed diagnoses at early screening conditions. Feature importance analysis also reports that important hormonal biomarkers like LH/FSH ratio, BMI, TSH and prolactin are the most significant factor in prediction of PCOS followed by life-style predictors such as physical activity levels and level of sleep quality. These results highlight the importance of consideration of hormonal and lifestyle indices clinico-biodata into analyses and confirm that CatBoost is a viable solution for dealing with heterogeneous healthcare data providing an accurate, interpretable model for non-invasive PCOS risk prediction.

VI Conclusion

In conclusion, this work has introduced an efficient framework for risk screening and personalized guidance in PCOS based on the CatBoost classifier by incorporating hormonal biomarkers and lifestyle health markers accurately without any invasive. The predictive performance of the model was high with accuracy, precision, recall and F1-score 96.3%,95.8%,96.9% and 96.3%, respectively, and widely confirms the robustness and reliability of the model to discriminate PCOS from non-PCOS cases (AUC=0.97). The intrinsic capability of CatBoost in processing categorical features well and mining deep complex nonlinear interdependent structures plays an important role in its excellent generalization and interpretability. By integrating risk prediction and individualized lifestyle advice, the proposed system facilitates the transition from computational diagnosis to concrete healthcare assistance, allowing for timely interventions and making well-informed decisions. Future work will involve validation and implementation of our proposed framework on larger more heterogeneous, multi-center data sets and incorporation of longitudinal as well as wearable sensor data to improve the real time monitoring and clinical efficacy in PCOS.

References

[1] Rahman, M. M., Islam, A., Islam, F., Zaman, M., Islam, M. R., Sakib, M. A., & Babu, H. H. (2024). Empowering early detection: A web-based machine learning approach for PCOS prediction. Informatics in Medicine Unlocked, 47, 101500.

[2] Aggarwal, S., & Pandey, K. (2023). Early identification of PCOS with commonly known diseases: obesity, diabetes, high blood pressure and heart disease using machine learning techniques. Expert Systems with Applications, 217, 119532.

[3] Panjwani, B., Yadav, J., Mohan, V., Agarwal, N., & Agarwal, S. (2025). Optimized Machine Learning for the Early Detection of Polycystic Ovary Syndrome in Women. Sensors, 25(4), 1166.

[4] Elmannai, H., El-Rashidy, N., Mashal, I., Alohali, M. A., Farag, S., El-Sappagh, S., & Saleh, H. (2023). Polycystic ovary syndrome detection machine learning model based on optimized feature selection and explainable artificial intelligence. Diagnostics, 13(8), 1506.
[5] Kodipalli, A., Devi, S., & Dasar, S. (2024). Semantic segmentation and classification of polycystic ovarian disease using attention UNet, Pyspark, and ensemble learning model. Expert Systems, 41(3), e13498.
[6]Ahmad, R., Maghrabi, L. A., Khaja, I. A., Maghrabi, L. A., & Ahmad, M. (2024). Smote-based automated pcos prediction using lightweight deep learning models. Diagnostics, 14(19), 2225. Vedavathi, N., Narender, M., Johnpaul, C. I., & Mathew, T. (2024). Polycystic Ovary Syndrome Detection Using Contextual Ensemble Network and ELMAN Neural Network with Green Anaconda Optimization. Journal of Multiscale Modelling, 15(04), 2450007.
[7] Faris, N., Sahi, A., Diykh, M., Abdulla, S., & Siuly, S. (2025). Enhanced Polycystic Ovary Syndrome diagnosis model leveraging a K-means based genetic algorithm and ensemble approach. Intelligence-Based Medicine, 100253.
[8] Khanna, V. V., Chadaga, K., Sampathila, N., Prabhu, S., Bhandage, V., & Hegde, G. K. (2023). A distinctive explainable machine learning framework for detection of polycystic ovary syndrome. Applied System Innovation, 6(2), 32.
[9] Jothimani, S., Sangeethaa, S. N., Balamurugan, M., & Kayalvili, S. (2025). Detection of Polycystic Ovarian Disease Using Probabilistic Optimum Deep Learning Feature Fusion. IETE Journal of Research, 1-15.
[10] Tiwari, S., Kane, L., Koundal, D., Jain, A., Alhudhaif, A., Polat, K., ... & Althubiti, S. A. (2022). SPOSDS: A smart Polycystic Ovary Syndrome diagnostic system using machine learning. Expert Systems with Applications, 203, 117592.
[11] Khan, I., & Khare, B. K. (2025). Diagnosis of polycystic ovary syndrome (PCOS) using TabNet and BiLSTM with attention model. Health Systems, 1-16.
[12] Rajput, I. S., Tyagi, S., Gupta, A., & Jain, V. (2024). Sine cosine algorithm-based feature selection for improved machine learning models in polycystic ovary syndrome diagnosis. Multimedia Tools and Applications, 83(30), 75007-75031.
[13] Capao, B. M. B., Guido Jr, C. A., & Peña, C. F. (2023, October). Performance Evaluation of Heterogeneous Ensemble Methods for Polycystic Ovarian Syndrome Detection. In International Conference on Intelligent Computing & Optimization (pp. 104-113). Cham: Springer Nature Switzerland.
[14] Sarwar, M. S., & Rehman, S. U. (2025). Hybrid quantum-classical machine learning for enhanced PCOS classification. Quantum Machine Intelligence, 7(2), 83.
[15] Gupta, D., Gupta, A., Sharma, S., Kumar, S., Rai, P., Kaur, K., & Jindal, N. (2025). PCOS and Gyno Help: AI based app and web application development. Journal of The Institution of Engineers (India): Series B, 106(5), 1455-1465.

[16] Aldosary, A., El-Shafai, W., Mohamed, E. M., & Emara, H. M. (2025). Enhancing PCOS diagnosis accuracy with modified Giza Pyramids Optimizer, SLOGPO, and PCOSNet: advanced classification and optimization techniques. Cluster Computing, 28(5), 1-27.
[17] Ahmed, S., Rahman, M. S., Jahan, I., Kaiser, M. S., Hosen, A. S., Ghimire, D., & Kim, S. H. (2023). A review on the detection techniques of polycystic ovary syndrome using machine learning. IEEE Access, 11, 86522-86543.
[18] Dhanka, S., Kumar, A., Sharma, A., Vundavilli, H., Maini, S., & Rajasekar, E. (2025). Advances in machine learning and deep learning for hormonal disorder diagnosis: an exhaustive review on PCOS, thyroid, and optimization techniques. Archives of Computational Methods in Engineering, 1-45.
[19] Govindharajan, G., Subramanian, S., Doraipandian, M., & Rajendran, S. (2025). Innovative ai-driven models for predicting polycystic ovarian syndrome: An extensive review of machine learning and deep learning frameworks. Archives of Computational Methods in Engineering, 1-26.
[20] Verma, P., Maan, P., Gautam, R., & Arora, T. (2024). Unveiling the role of artificial intelligence (AI) in polycystic ovary syndrome (PCOS) diagnosis: A comprehensive review. Reproductive Sciences, 31(10), 2901-2915.
[21] Vairachilai, S., Anuhya, D., Tirkey, A., & Raja, S. P. (2024). SLB-SMOTE logistic blending hybrid machine learning model for chronic polycystic ovary syndrome prediction with correlated feature selection. Informatics for Health and Social Care, 49(3-4), 190-211.
[22] Ghaderzadeh, M., Garavand, A., & Salehnasab, C. (2025). Artificial intelligence in polycystic ovary syndrome: A systematic review of diagnostic and predictive applications. BMC Medical Informatics and Decision Making, 25(1), 427.
[23] Ramteke, M., & Raut, S. (2025). Prediction of polycystic ovary syndrome using machine learning with SFS and Boruta feature selection: an explainable AI approach. Systems Biology in Reproductive Medicine, 71(1), 439-460.
[24] Krishana, S., Sarvadeo, S. S., Sharma, S., & Sharma, S. (2025). A Multi‐Technique Feature Selection Framework Using Computational Intelligence and Internet of Medical Things for Early Detection of Polycystic Ovary Syndrome in Women. Computational Intelligence, 41(4), e70122.
[25] Arora, S., Vedpal, & Chauhan, N. (2025). Polycystic Ovary Syndrome (PCOS) diagnostic methods in machine learning: a systematic literature review. Multimedia Tools and Applications, 84(16), 16301-16337.
[26]https://www.kaggle.com/datasets/prasoonkottarathil/polycystic-ovary-syndrome-pcos